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WEB TECHNOLOGIES

HTML 5 – Geo Location

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Acknowledgement:

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- Enables your web application to obtain the geographical position of your website visitors
- The user has to accept to share their location,
- The Geolocation API is work with the `navigator.geolocation` object.
- Accessed via JavaScript, through the `navigator.geolocation` object.
- `navigator.geolocation` object allows you to access geo location through two primary functions:
 - `getCurrentPosition()`
 - Returns the location of the visitor as a one-time snapshot
 - `watchPosition()`
 - returns the location of the visitor every time the location changes

Syntax : `getCurrentPosition(success, error, options)`
`watchPosition(success, error, options)`

- Both functions take the following parameters:
 - Success callback function
 - Error callback function (optional)
 - Geo location options object (optional)
 - If the `getCurrentPosition()` method is successful, it returns a coordinates object to the function specified in the parameter (success)

HTML5 – Geo Location

Introduction

The Geolocation API uses three methods of Geolocation interface which are given following:

Methods	Description
<code>getCurrentPosition()</code>	It identifies the device or the user's current location and returns a position object with data.
<code>watchPosition()</code>	Return a value whenever the device location changes.
<code>clearWatch()</code>	It cancels the previous <code>watchPosition()</code> call

```
navigator.geolocation.getCurrentPosition(showPosition);  
function showPosition(position) {  
    x.innerHTML = "Latitude: " + position.coords.latitude +  
        "<br>Longitude: " + position.coords.longitude;  
}
```

```
navigator.geolocation.getCurrentPosition( function(position) {  
    console.log("your position is: " + position.coords.latitude + ", " +  
        position.coords.longitude);  
} );
```

* `getCurrentPosition` can be replaced with `watchPosition` if you need to receive updates to location

1. Which JavaScript object provides access to the HTML5 Geolocation API?

- A) window.location
- B) navigator.location
- C) navigator.geolocation
- D) document.geolocation

2. Which Geolocation API method returns the user's location as a one-time snapshot?

- A) getPosition()
- B) getCurrentPosition()
- C) watchPosition()
- D) trackPosition()

3. In the Geolocation API, the first parameter of getCurrentPosition() is:

- A) Error callback function
- B) Success callback function
- C) Options object
- D) Coordinates object

4. What does the position object returned by Geolocation API contain?

- A) Only latitude
- B) Only longitude
- C) Both latitude and longitude in coords property
- D) URL of user's location

5. If the user denies location permission, which callback is executed in the `getCurrentPosition()` method?

`navigator.geolocation.getCurrentPosition(successCallback, errorCallback);`

- A) successCallback
- B) errorCallback
- C) Both callbacks execute simultaneously
- D) None of them

Answer 1: C) navigator.geolocation

Answer 2: B) getCurrentPosition()

Answer 3: B) Success callback function

Answer 4: C) Both latitude and longitude in coords property

Answer 5: B) errorCallback



THANK YOU

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